Database Concepts – Key Answers

1. SQL vs NoSQL (Key Differences)

Feature	SQL	NoSQL
Structure	Tables (rows & columns)	Document / Key-Value / Graph
Schema	Fixed	Flexible
Scalability	Vertical	Horizontal
Consistency	ACID	BASE (eventual consistency)
Examples	MySQL, PostgreSQL	MongoDB, Cassandra

2. CAP Theorem

A distributed system can only guarantee two of the following three properties:

Consistency: All nodes see the same data.

Availability: Every request gets a response.

Partition Tolerance: System works despite network failures.

During a partition, a system must choose between consistency and availability.

3. When to Prefer MongoDB

• Flexible schema (e.g., user profiles with varying fields)

• Large-scale applications (e.g., analytics, logs)

• JSON-like data storage (e.g., APIs, microservices)

4. Why MongoDB Uses BSON

BSON is a binary format that is faster and more efficient than JSON.

Supports additional data types like Date, Binary, ObjectId.

Improves storage efficiency and indexing performance.

5. MongoDB Query Example

Find students with GPA > 3.5 and enrolled in CS101:

```
db.students.find({ gpa: { $gt: 3.5 }, course: 'CS101' })
```